Named Entity Recognition – South African Chrome & Platinum Mining Industry

Task type: Named Entity Recognition (span-based text tagging)

Tool: Label Studio

Source text: Custom-authored corpus covering South Africa's chrome and platinum mining sector

Status: Complete

Overview

This project applies named entity recognition to a domain-specific text corpus covering South Africa's chrome and platinum mining industry, including corporate announcements, production figures, safety incidents, and international trade agreements.

It extends a standard 6-class NER schema (PER, ORG, LOC, DATE, TIME, MISC) with four additional domain-specific classes — **MONEY, PERCENT, MINERAL, and FACILITY** — added specifically to capture the kinds of entities that recur in industrial/mining text but aren't covered by a general-purpose schema. A further **QUANTITY** class was added mid-project after identifying a gap in the original schema (production figures like "42,000 ounces" didn't fit any existing category), demonstrating iterative schema refinement in response to real data.

This is the second project in an ongoing NER series, building on an earlier general-purpose NER sample and showing progression toward more specialized, industry-specific entity extraction.

Entity Schema

Tag	Label	Definition
PER	Person	Named individuals
ORG	Organization	Companies, unions, regulatory bodies, government departments

LOC	Location	Cities, provinces, countries, regions
DATE	Date	Calendar dates, fiscal year references (day-of-week merged into the same span as the full date)
TIME	Time	Clock times
MONEY	Monetary value	Currency amounts in any denomination
PERCENT	Percentage	Figures expressed as a percentage
MINERAL	Mineral/Commodity	Named minerals, metals, or refined commodities
FACILITY	Facility/Site	Named mines, shafts, or plants — distinct from the company that owns/operates them
QUANTITY	Quantity	Numeric production/output measurements paired with a unit (added mid-project to close a schema gap)
MISC	Miscellaneous	Legislation, stock tickers, contact identifiers (email/phone), and other named entities not covered above

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Key Judgment Calls & Rules Applied

- **ORG vs. FACILITY:** the operating company (e.g., "Anglo American Platinum," "Impala Platinum") is tagged ORG, while the physical mine/shaft/site it operates (e.g., "Mogalakwena Mine," "Merensky Shaft") is tagged FACILITY as a separate entity, even when they appear in the same phrase.
- **Stock tickers** (e.g., "NPH") are tagged MISC, not ORG, since they denote a listing symbol rather than a distinct organization.
- **Combined date spans:** a day-of-week appearing directly alongside a full date (e.g., "Monday, 14 September 2026") is tagged as a single DATE entity rather than split into two.
- **Combined location spans:** city+region/country pairs (e.g., "Lydenburg, Mpumalanga") are tagged as a single LOC entity.
- **Possessives:** possessive markers ('s) are excluded from entity spans for consistency (e.g., "Glencore" not "Glencore's").
- **Contact identifiers** (email addresses, phone numbers) are tagged MISC, consistent with the rule established in the first NER project in this series.

Revision Process

An initial annotation pass surfaced several errors during review, including a company misclassified as a MINERAL, a stock ticker misclassified as an ORG, and a facility/company role mix-up. These were identified, corrected in a targeted second pass, and re-reviewed rather than reworked from scratch — reflecting a realistic professional QA workflow rather than a single-pass "perfect on the first try" approach.

Files in This Submission

- `README.md` — this file
- `entity_schema.md` — full entity definitions and labeling rules
- `annotations_export.json` — raw exported annotations from Label Studio
- `screenshots/` — sample tagged passages showing the finalized annotation

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Reflection

This project demonstrates the ability to extend a general NER schema to fit a specialized domain, identify and close schema gaps during annotation rather than forcing data into ill-fitting categories, and self-correct through a structured revision process — skills directly relevant to production-quality training data work in specialized industries.